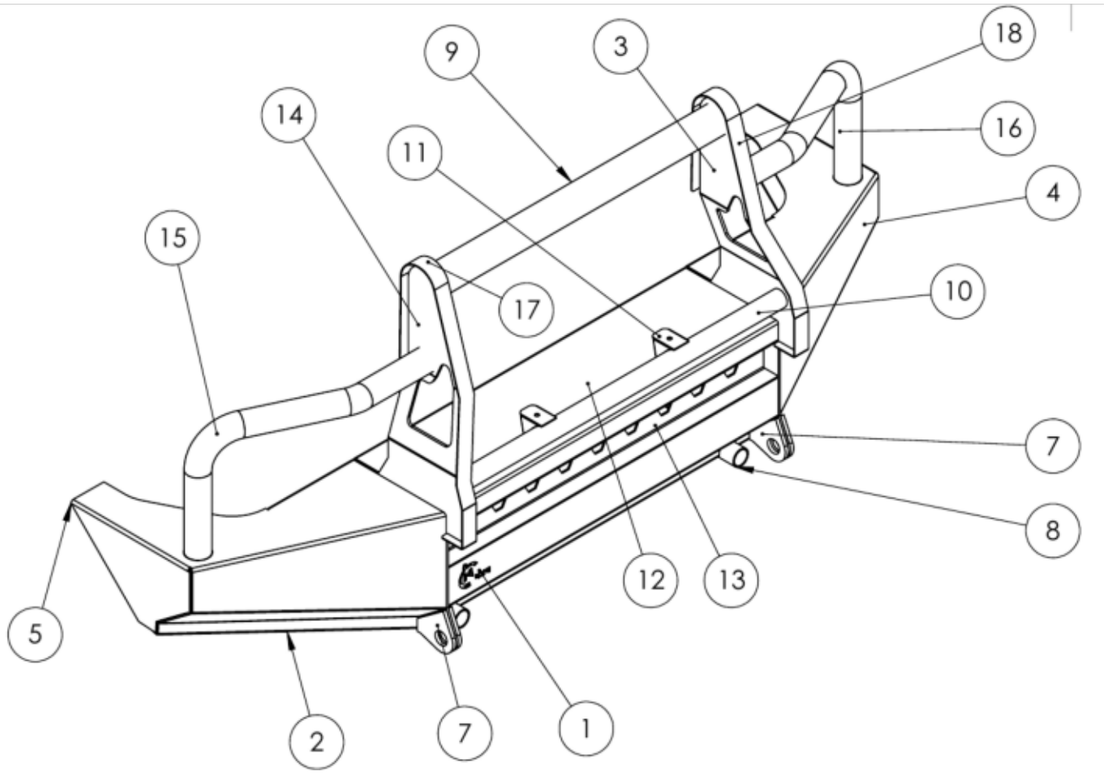
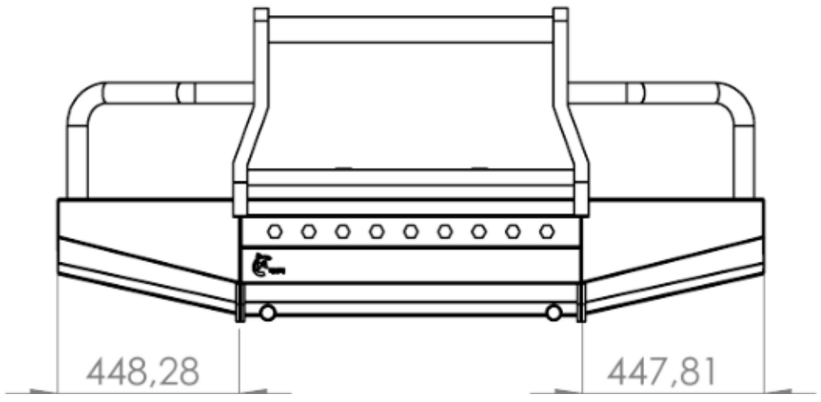
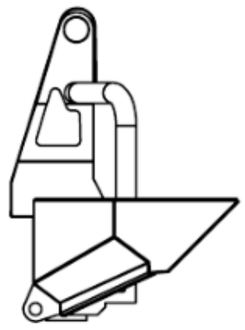
ASSEMBLY - item balloons per source BOM (see table)



FRONT / SIDE - overall 1741 mm; centre 835 mm; ends 448 mm





BILL OF MATERIALS / CUT LIST

Item	VA part no.	Description	Thk	Qty	Flat (mm)	kg ea
1	VA-BB-01	Centre panel	3	1	835 × 209	4.05
2	VA-BB-02	End panel (item 2)	3	1	688 × 687	6.32
3	VA-BB-03	Upright RH	5	1	306 × 769	4.66
4	VA-BB-04	End panel, opposite hand (item 4)	3	1	688 × 687	6.32
5	VA-BB-05	End plate	3	2	269 × 78	0.50
6	VA-BB-06	Bracket 1 (chassis bracket)	5	2	184 × 258	1.82
7	VA-BB-07	Recovery lug	12	4	91 × 92	0.44
8	VA-BB-08	Pipe lift (jacking stub)	38.1 × 2.0	2	L = 100	0.18
9	VA-BB-09	Pipe centre (top hoop)	63.5 × 2.0	1	L = 702	2.15
10	VA-BB-10	Pipe spot (light bar)	38.1 × 2.0	1	L = 805	1.45
11	VA-BB-11	Spot bracket (light mount)	2	2	60 × 89	0.06
12	VA-BB-12	Centre panel P2 (top plate)	3	1	835 × 289	5.73
13	VA-BB-13	Centre panel P3 (lower channel)	3	1	835 × 153	2.87
14	VA-BB-14	Upright LH	5	1	306 × 769	4.66
15	VA-BB-15	Tube 50 LH (side hoop)	50 × 2.0	1	L = 510 (bent)	1.22
16	VA-BB-16	Tube 50 RH (side hoop)	50 × 2.0	1	L = 510 (bent)	1.22
17	VA-BB-17	Face 1 (RH trim strip)	2	1	860 × 87	0.48
18	VA-BB-18	Face 1 LH (trim strip)	2	1	860 × 87	0.48

NEST GROUPS BY THICKNESS (laser)

2 mm 304 SS: 11 ×2, 17 ×1, 18 ×1 | net 0.07 m2 | 1.1 kg

3 mm 304 SS: 01 ×1, 02 ×1, 04 ×1, 05 ×2, 12 ×1, 13 ×1 | net 1.11 m2 | 26.3 kg

5 mm 304 SS: 03 ×1, 06 ×2, 14 ×1 | net 0.33 m2 | 13.0 kg

12 mm 304 SS: 07 ×4 | net 0.02 m2 | 1.8 kg

PROCESS NOTES

1. Laser cut all plate 1:1 from DXF (layer CUT). Layers BEND_UP / BEND_DOWN are bend-line centres - do not cut.
2. Layer ETCH = low-power surface mark (wordmark on centre panel). Layer INFO = reference text only.
3. Flats are developed at K = 0.46 (air bend, 304). 304 springs back 2-3 deg more than mild steel - over-bend and check first-off against the bend table.
4. Laser cut with nitrogen (oxide-free edge). Deburr; break sharp corners 0.5 mm. Stainless-only tooling, films and brushes - no carbon-steel contamination. 12 mm lugs: plasma or 6 kW+ fibre laser.
5. Bend direction is viewed from the flat-pattern face shown on each sheet (UP = flange toward viewer).
6. Tube items 8/9/10/15/16: cut lengths and bend data on the TUBE SCHEDULE sheet, measured from the 1:20 assembly views. Wall 2.0 mm on all tubes.
7. Items 17/18 (face strips): bend as drawn (the source "NO BENDING" note is superseded). The R42 wrap matches the upright top radius.
8. Weld sequence: tack brackets (06) to centre (01) and P2/P3 (12/13); fit uprights (03/14) and lugs (07); add ends (02/04) and end plates (05) last.
9. Several source bend radii are below 1T (red warnings on sheets 05, 06, 11, 03/14, 17/18). Workable in 304 with a sharp-nose punch; confirm on first-off - if it cracks, open the radius.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-ASM

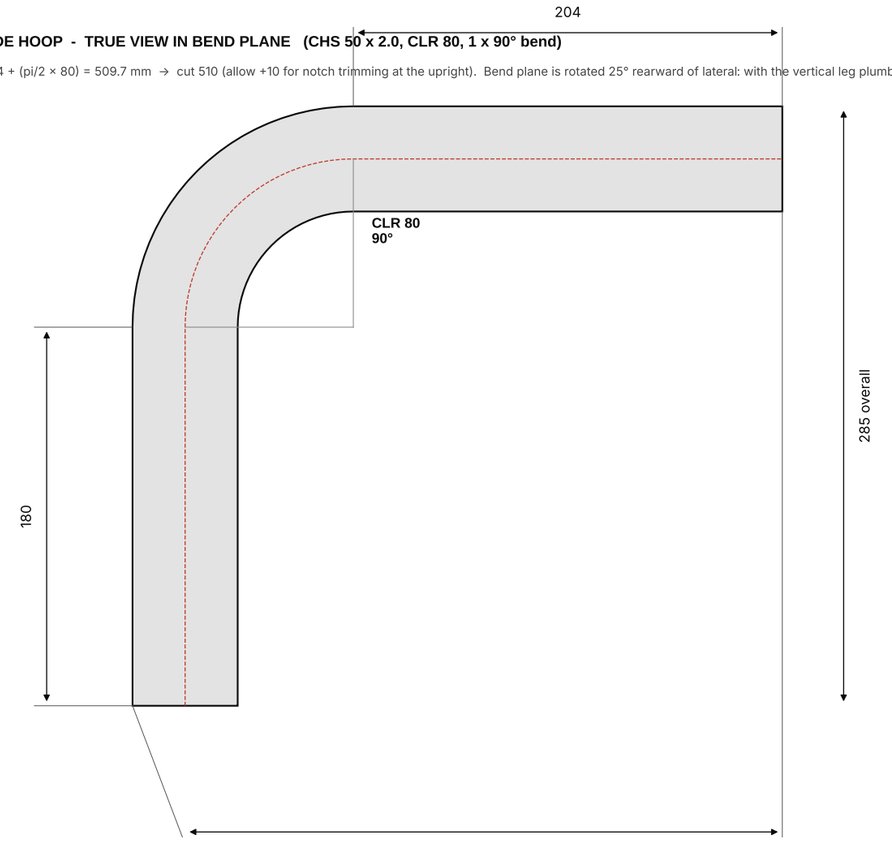
TITLE
BULL BAR - ASSEMBLY, BOM & CUT LIST

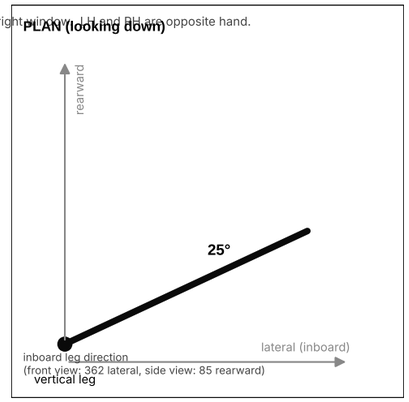
MATERIAL **304 SS 2B**	SHEET PARTS **19**	PLATE / TUBE MASS **42.1 / 6.4 kg**
OVERALL WIDTH 1741 mm	CENTRE SECTION 835 mm	END SECTIONS 448 mm each
THICKNESSES 2 / 3 / 5 / 12 mm	FITMENT TBC (per source 4×4 design)	TUBE CHS 50 / 63.5 / 38.1
WELDING TIG/MIG, ER308L, purge	FINISH Brushed, passivated	PACK VA-BB rev A

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

VA-BB-15 / 16 SIDE HOOP - TRUE VIEW IN BEND PLANE (CHS 50 x 2.0, CLR 80, 1 x 90° bend)

Cut length = 180 + 204 + (pi/2 × 80) = 509.7 mm → cut 510 (allow +10 for notch trimming at the upright). Bend plane is rotated 25° rearward of lateral: with the vertical leg plumb, the inboard leg points toward the upright window. LH and RH are opposite hand.





TUBE SCHEDULE

VA-BB-08 Pipe lift (jacking stub) x2

CHS 38.1 × 2.0 cut length 100 mm 0.18 kg ea [approx.]

Seen end-on under the centre section in the front view (Ø38 outer / Ø32 inner at 705 mm centres); length from side view.

VA-BB-09 Pipe centre (top hoop) x1

CHS 63.5 × 2.0 cut length 702 mm 2.15 kg ea [measured]

Straight tube between the upright faces at the top (front view 5529.4 to 6231.4); Ø63 end-on in side view.

VA-BB-10 Pipe spot (light bar) x1

CHS 38.1 × 2.0 cut length 805 mm 1.45 kg ea [measured]

Straight tube between the upright feet (front view 5477.9 to 6282.9), 38.1 wide in front view.

VA-BB-15 Tube 50 LH (side hoop) x1

CHS 50 × 2.0 cut length 510 mm 1.22 kg ea [measured]

bend 90°, CLR 80, legs 180 / 204 (tangent to end), bend plane 25° rearward

Vertical leg 180 from end-panel top to tangent; 90 deg bend, CLR 80 (105 outer / 55 inner); inboard leg 204 to end, running 25 deg rearward of lateral into the upright window.

VA-BB-16 Tube 50 RH (side hoop) x1

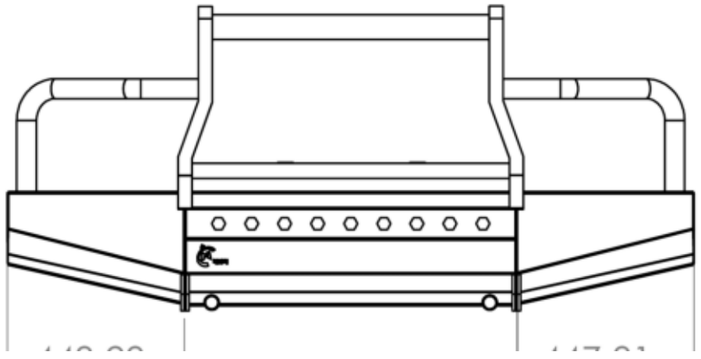
CHS 50 × 2.0 cut length 510 mm 1.22 kg ea [measured]

bend 90°, CLR 80, legs 180 / 204 (tangent to end), bend plane 25° rearward

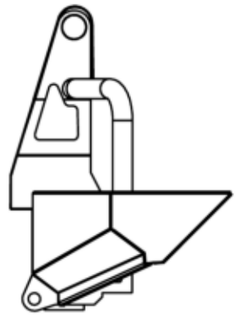
Mirror of VA-BB-15.

Measured from source assembly views at 1:20 (scale set from the 1741.09 dim). Wall 2.0 mm on all tubes per Veld Armour. Verify first-off on the jig.

SOURCE FRONT VIEW (measured)



SOURCE SIDE VIEW (measured)



HOW THE LENGTHS WERE TAKEN

Vector geometry extracted from the assembly PDF; the 1741.09 overall width fixes the scale. Hoop: vertical-leg tangent and horizontal-leg tangent found from the tube silhouettes in both views; CLR = 80 from the tangent offsets (72 lateral, 33 rearward, 80 vertical). Inboard leg true length from its two projections (184 lateral / 85 rearward). Centre and spot tubes span between the upright faces/feet. Lift stubs: Ø38/Ø32 circles under the bar, length from the side view.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.

VA-BB-TUBE

TITLE

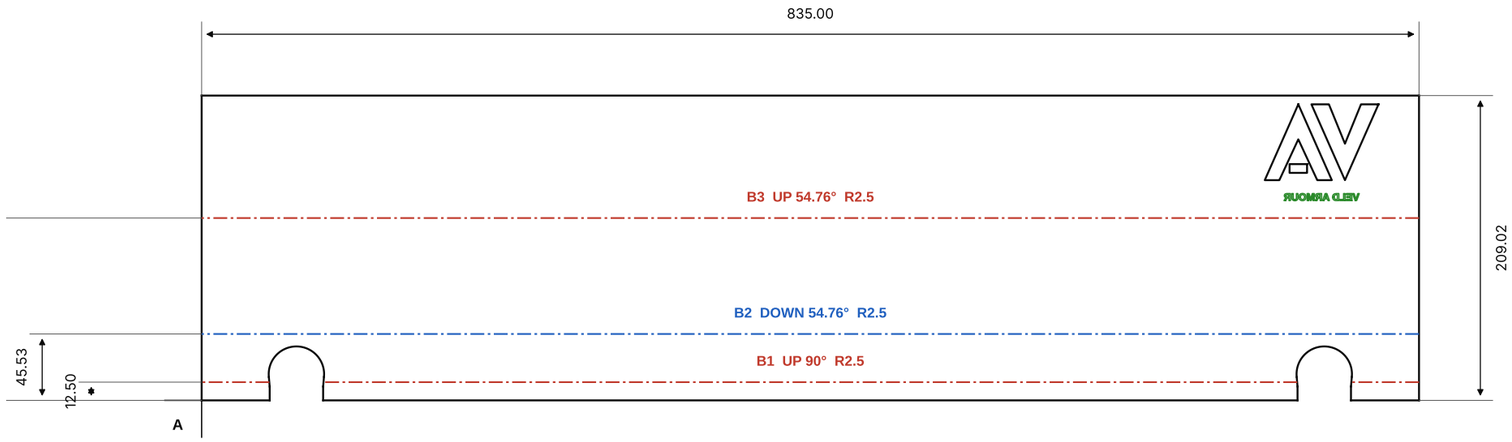
TUBE SCHEDULE - CUT LENGTHS & BEND DATA

MATERIAL **304 SS CHS**	TUBE ITEMS **6**	TUBE MASS **6.4 kg**
HOOP CLR 80 mm (1.6 D)	BEND METHOD Mandrel / rotary draw	SOURCE SCALE 1:20, sheet 1
WALL 2.0 mm all tubes	FITMENT TBC	END CUTS square unless noted
TOL. LENGTH +/- 1 mm	TOL. ANGLE +/- 1 deg	FINISH Brushed, passivated

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

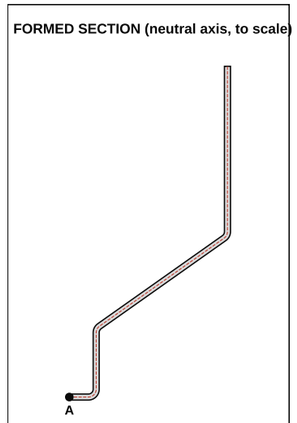


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2.5	12.50 (Y)	6.09	4.91	24
B2	DOWN	54.76°	2.5	45.53 (Y)	3.71	1.99	24
B3	UP	54.76°	2.5	125.04 (Y)	3.71	1.99	24

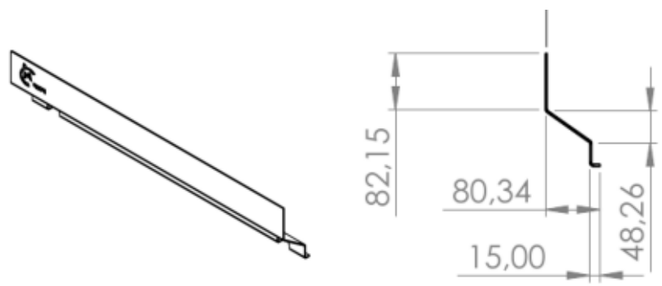
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 12.5 / 33.0 / 79.5 / 84.0 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Veld Armour monogram cut through; wordmark on ETCH layer (mirrored in flat, reads correct on outside face). (2) Half-round notches clear the upright lugs - check fit-up before final bend.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-01

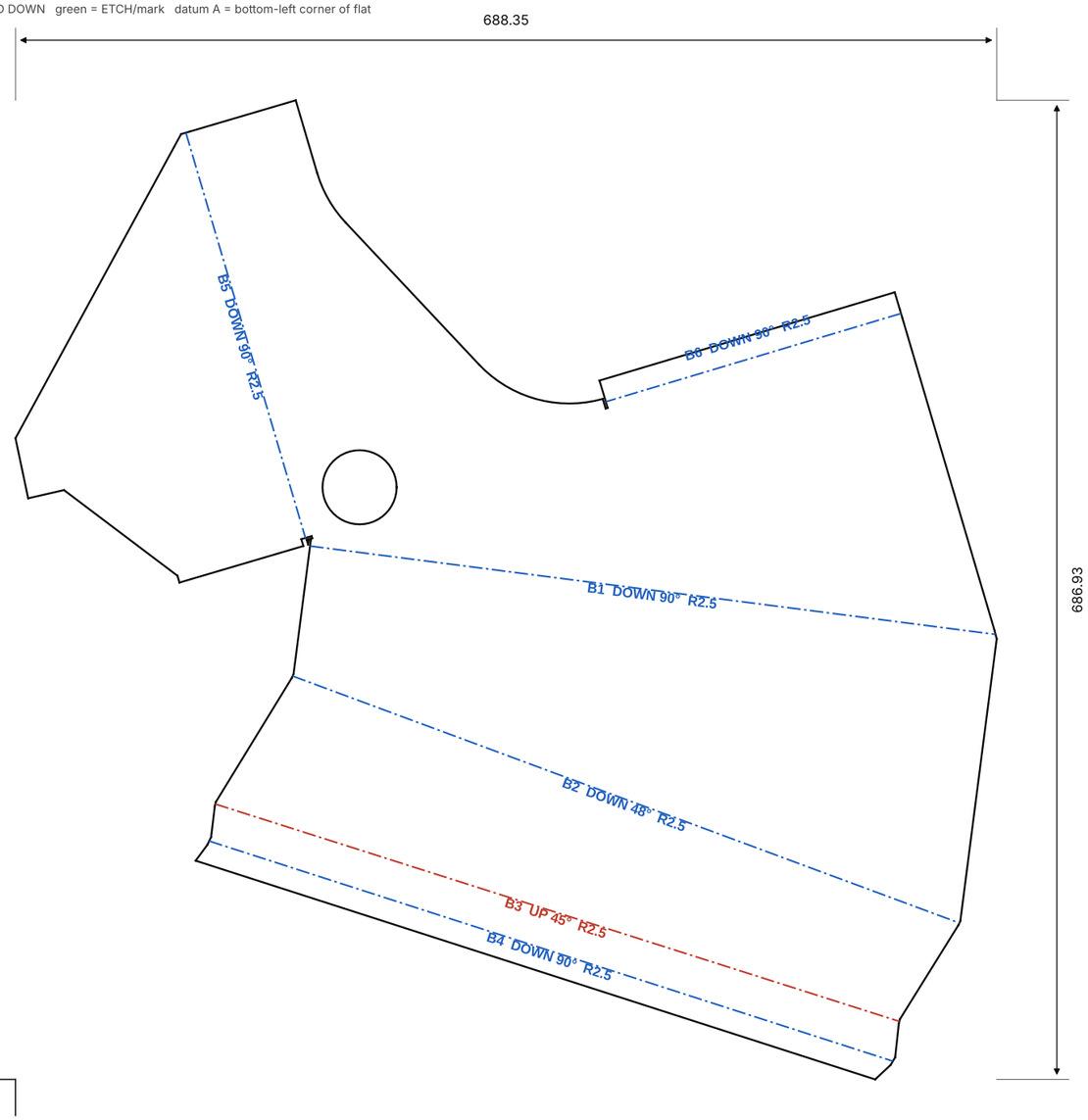
TITLE
Centre panel - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**3 mm**	**1**
FLAT SIZE	NET MASS	K-FACTOR
835.0 x 209.0	4.05 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
3	#1	VA-BB-01_Centre_panel_3mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

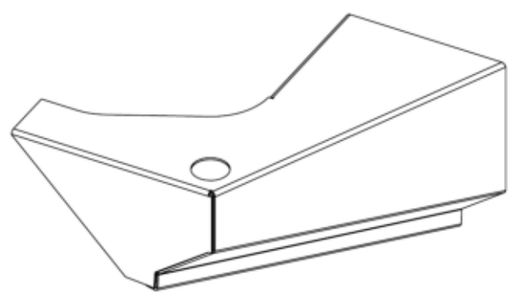
No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	2.5	see view	6.09	4.91	24
B2	DOWN	48°	2.5	see view	3.25	1.65	24
B3	UP	45°	2.5	see view	3.05	1.51	24
B4	DOWN	90°	2.5	see view	6.09	4.91	24
B5	DOWN	90°	2.5	see view	6.09	4.91	24
B6	DOWN	90°	2.5	see view	6.09	4.91	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3, B4, B5, B6 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)

Non-parallel bends - see formed view and assembly sheet for orientation.

FORMED VIEW (reference, not to scale)



NOTES: (1) Non-parallel bends: form the short 90 deg corner flanges last, after the 48/45/90 return lip. (2) Suggested sequence: B2 (48 deg) > B3 (45 deg up) > B4 (90 deg lip) > B1 (long 90) > B5, B6 (corner flanges).

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-02

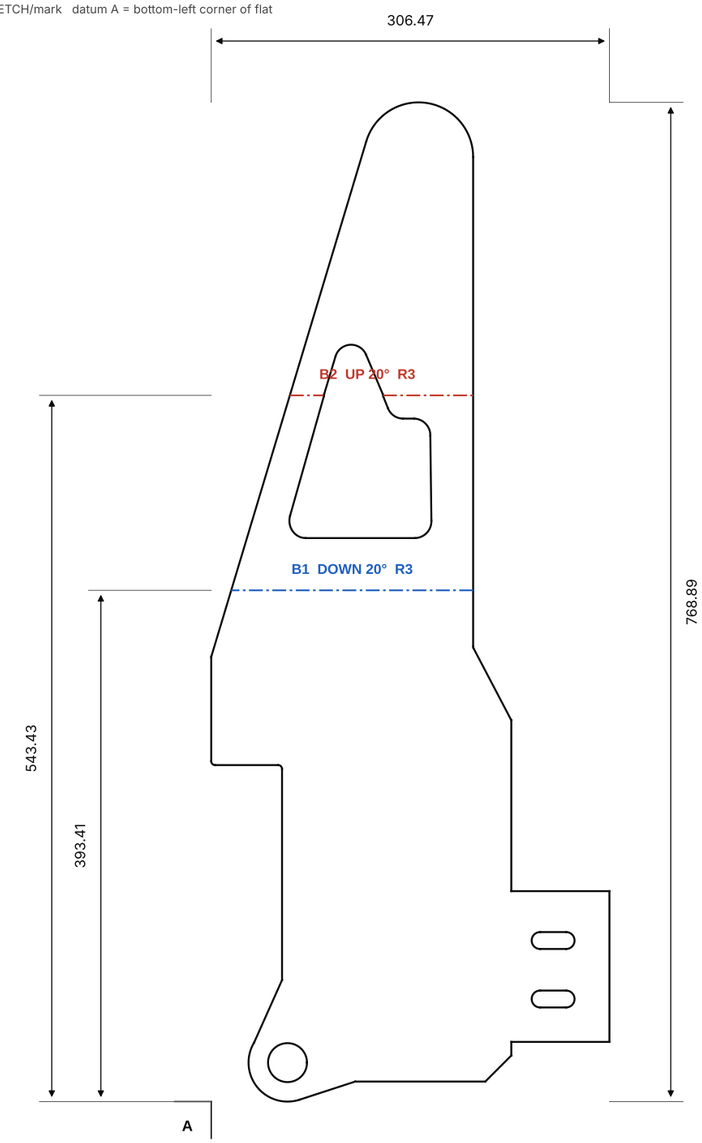
TITLE
End panel (item 2) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 688.3 × 686.9	NET MASS 6.32 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 6	SOURCE ITEM #2	DXF VA-BB-02_End_panel_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	20°	3	393.41 (Y)	1.85	0.97	40
B2	UP	20°	3	543.43 (Y)	1.85	0.97	40

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 393.4 / 150.0 / 225.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Two opposed 20 deg bends form a joggle - set back gauge from the slotted foot (datum edge A). (2) Lug (VA-BB-07) welds over the dia 30 hole; R30 boss on this part matches lug outer radius.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-03

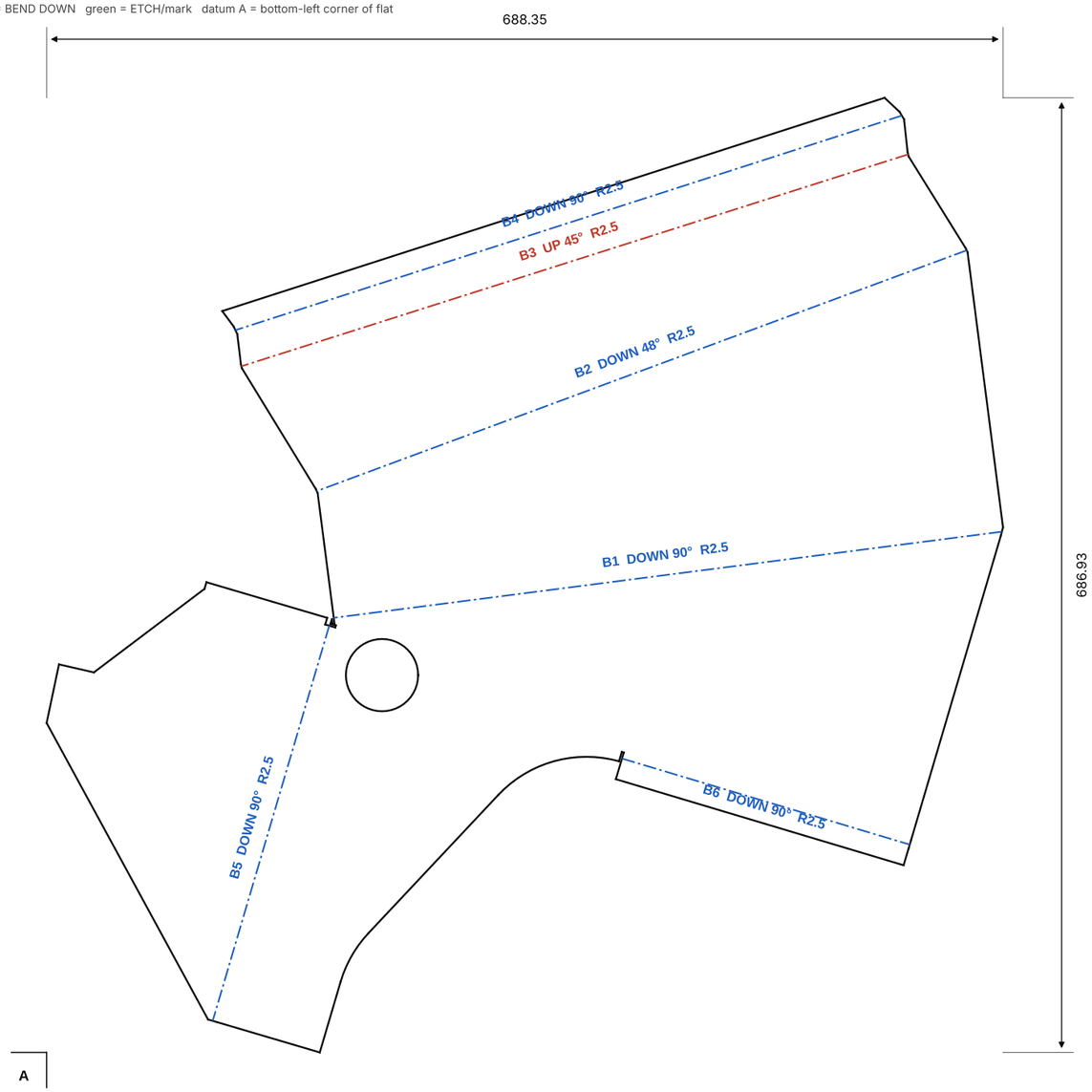
TITLE
Upright RH - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **5 mm**	QTY / ASSY **1**
FLAT SIZE 306.5 × 768.9	NET MASS 4.66 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #3	DXF VA-BB-03_Upright_RH_5mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

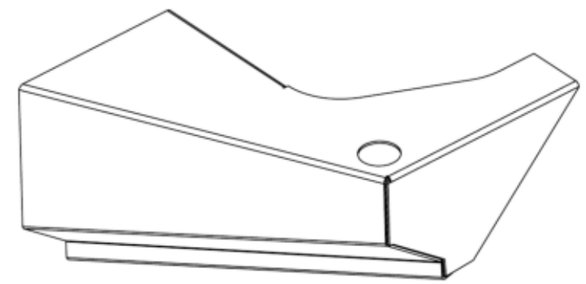
No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	2.5	see view	6.09	4.91	24
B2	DOWN	48°	2.5	see view	3.25	1.65	24
B3	UP	45°	2.5	see view	3.05	1.51	24
B4	DOWN	90°	2.5	see view	6.09	4.91	24
B5	DOWN	90°	2.5	see view	6.09	4.91	24
B6	DOWN	90°	2.5	see view	6.09	4.91	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2, B3, B4, B5, B6 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)

Non-parallel bends - see formed view and assembly sheet for orientation.

FORMED VIEW (reference, not to scale)



NOTES: (1) Mirror of VA-BB-02. Same bend sequence as VA-BB-02.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-04

TITLE
End panel, opposite hand (item 4) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 688.3 × 686.9	NET MASS 6.32 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 6	SOURCE ITEM #4	DXF VA-BB-04_End_panel,_opposite_hand_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:2

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	40.84°	1.5	243.32 (X)	2.05	1.30	24

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 243.3 / 25.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Single shallow bend; use a sharp-radius V-die (V = 6T = 18 mm).

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-05

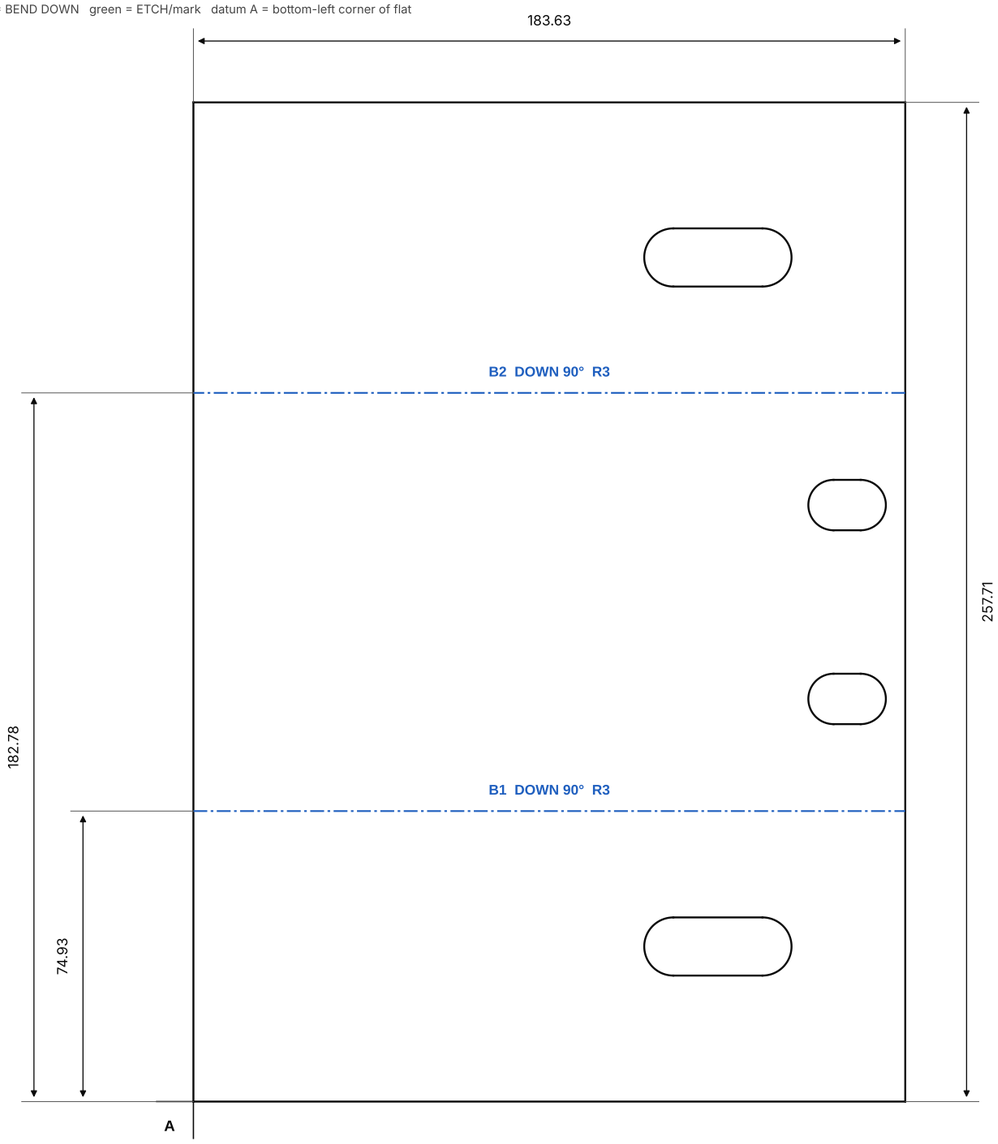
TITLE
End plate - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**3 mm**	**2**
FLAT SIZE	NET MASS	K-FACTOR
268.8 × 78.0	0.50 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
1	#5	VA-BB-05_End_plate_3mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:2

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

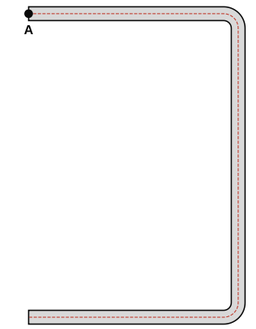


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	3	74.93 (Y)	8.33	7.67	40
B2	DOWN	90°	3	182.78 (Y)	8.33	7.67	40

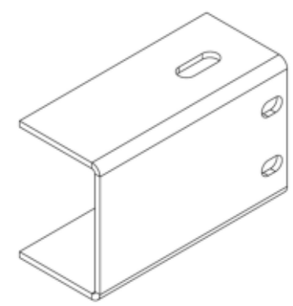
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 74.9 / 107.9 / 74.9 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) U-channel: bend both legs in the same direction. Check slot alignment to chassis rail after forming.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-06

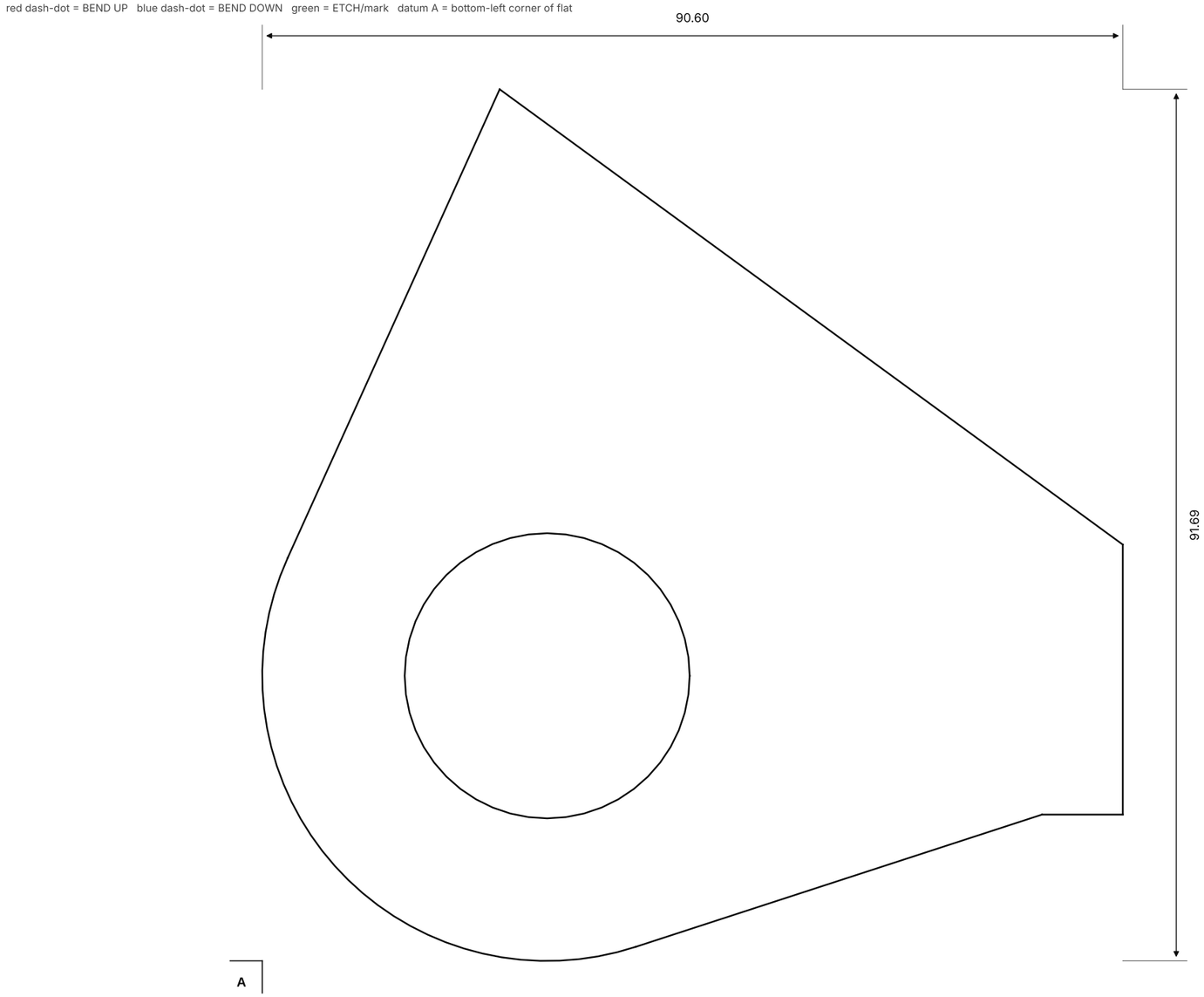
TITLE
Bracket 1 (chassis bracket) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**5 mm**	**2**
FLAT SIZE	NET MASS	K-FACTOR
183.6 × 257.7	1.82 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
2	#6	VA-BB-06_Bracket_1_5mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:1

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

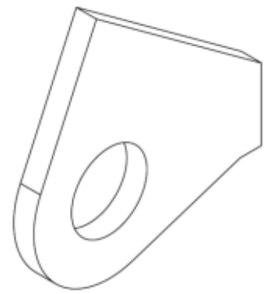
No	Dir	Angle	R in	From A	BA	BD	V-die
No bends - flat laser-cut part.							

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.

FORMED SECTION (neutral axis, to scale)

Flat part - no section.

FORMED VIEW (reference, not to scale)



NOTES: (1) Redrawn from assembly drawing sheet 8 (no flat supplied in source). Dia 30 hole, R30 boss. Upgraded from 8 mm to 12 mm (Rev A). (2) Laser cut 12 mm with nitrogen; dress the hole for a 30 mm pin / shackle bow. Weld through to the chassis bracket, not only to the 3 mm end panel.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-07

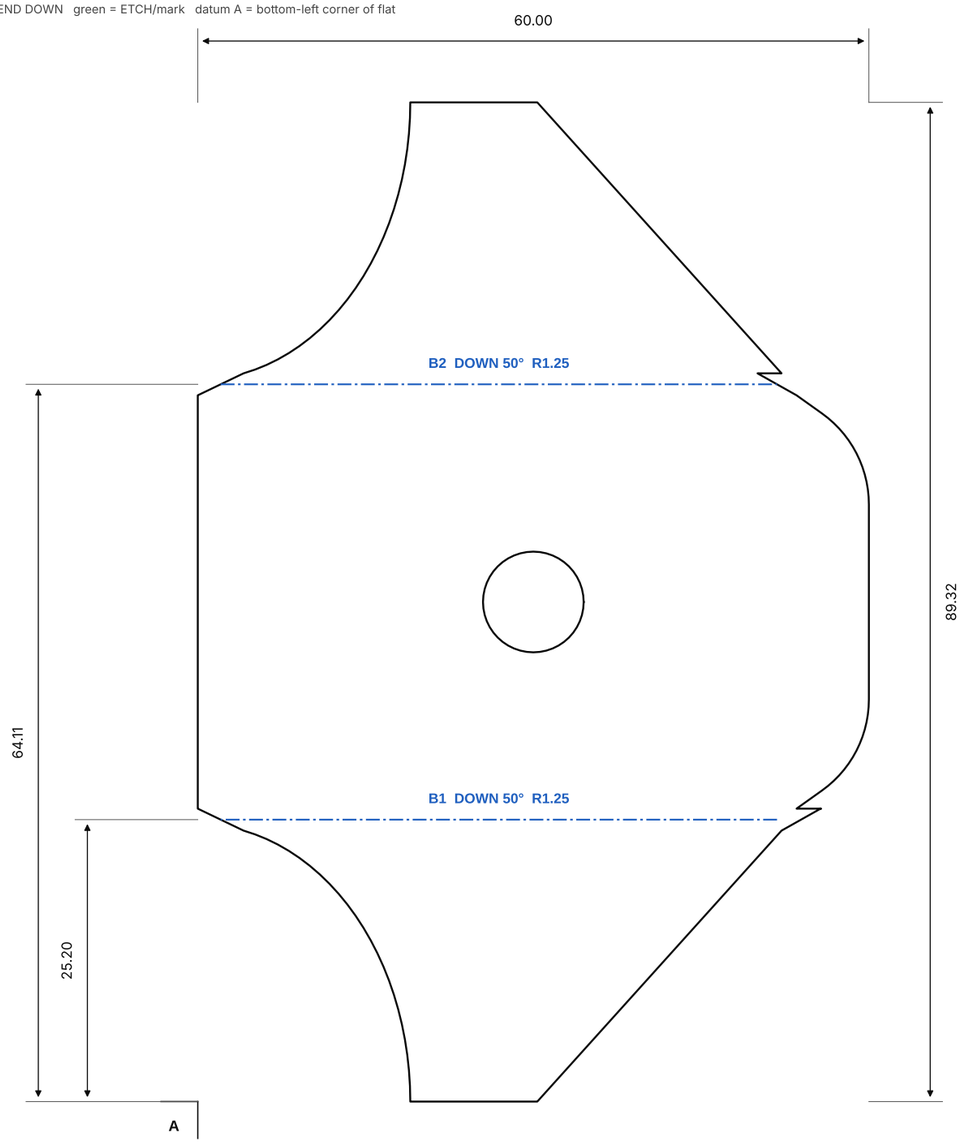
TITLE
Recovery lug - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**12 mm**	**4**
FLAT SIZE	NET MASS	K-FACTOR
90.6 × 91.7	0.44 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
0	#7	VA-BB-07_Recovery_lug_12mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:1

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

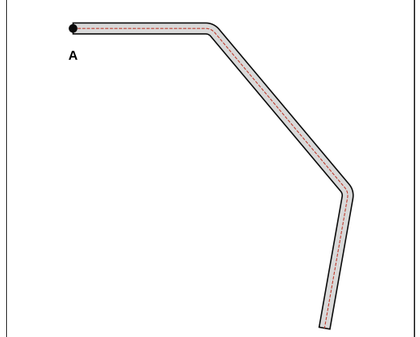


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	50°	1.25	25.20 (Y)	1.89	1.14	16
B2	DOWN	50°	1.25	64.11 (Y)	1.89	1.14	16

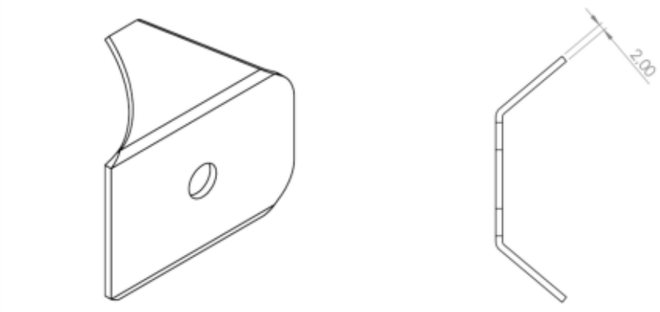
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 25.2 / 38.9 / 25.2 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Both wings bend the same way (down) to form a saddle. Dia 9 hole for M8 spot-light stud.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-11

TITLE
Spot bracket (light mount) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **2 mm**	QTY / ASSY **2**
FLAT SIZE 60.0 x 89.3	NET MASS 0.06 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #11	DXF VA-BB-11_Spot_bracket_2mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

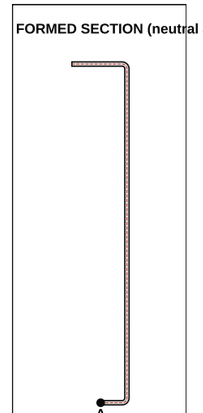


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2	17.51 (Y)	5.31	4.69	24
B2	UP	90°	2	251.18 (Y)	5.31	4.69	24

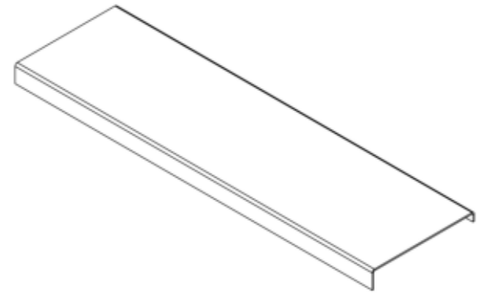
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 17.5 / 233.7 / 37.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Hat/channel section with unequal legs (17.5 and 37.5 mm). Short leg first.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-12

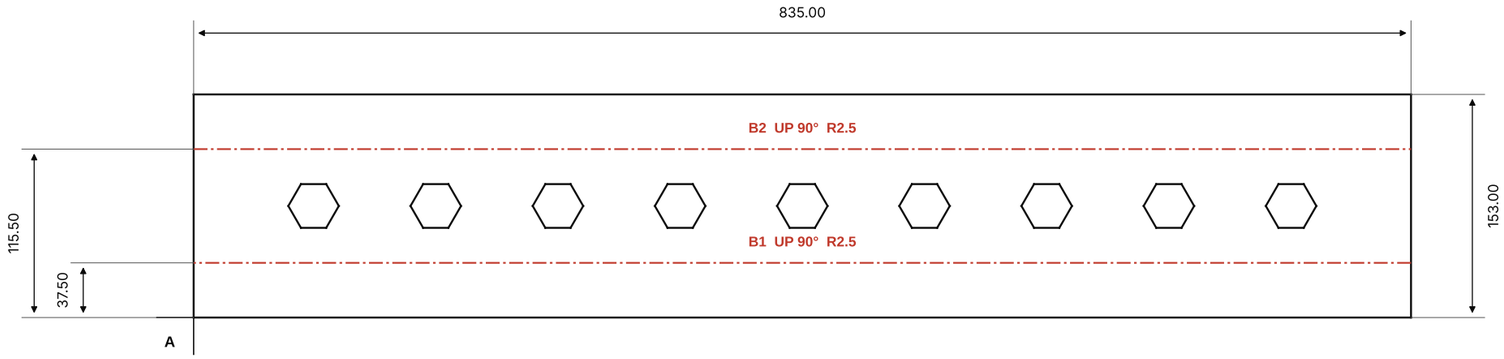
TITLE
Centre panel P2 (top plate) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**3 mm**	**1**
FLAT SIZE	NET MASS	K-FACTOR
835.0 x 288.7	5.73 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
2	#12	VA-BB-12_Centre_panel_P2_3mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	UP	90°	2.5	37.50 (Y)	6.09	4.91	24
B2	UP	90°	2.5	115.50 (Y)	6.09	4.91	24

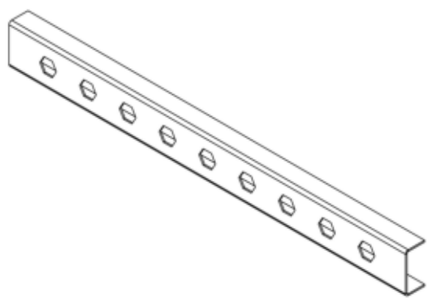
BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 3 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 37.5 / 78.0 / 37.5 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) 9 x hex apertures (20 mm A/F) on the web - decorative/vent. Equal 37.5 mm legs.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-13

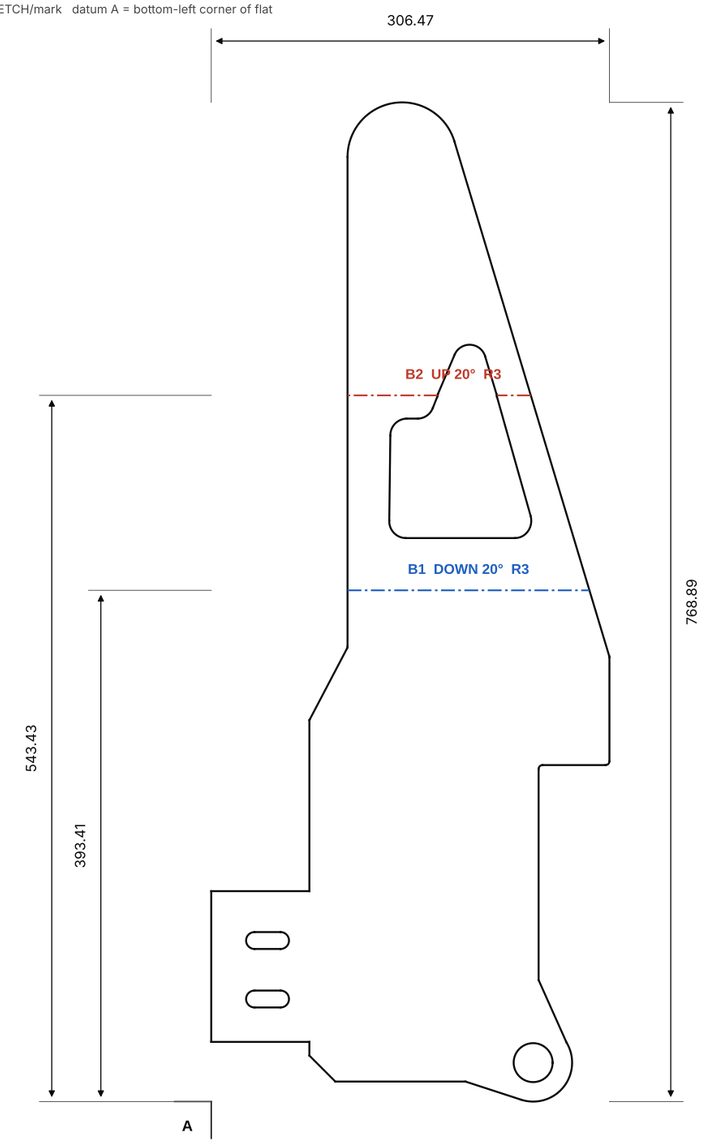
TITLE
Centre panel P3 (lower channel) - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **3 mm**	QTY / ASSY **1**
FLAT SIZE 835.0 x 153.0	NET MASS 2.87 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #13	DXF VA-BB-13_Centre_panel_P3_3mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat

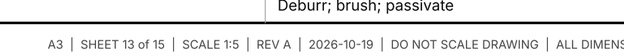


BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	20°	3	393.41 (Y)	1.85	0.97	40
B2	UP	20°	3	543.43 (Y)	1.85	0.97	40

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 5 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 393.4 / 150.0 / 225.5 mm

FORMED VIEW (reference, not to scale)



opposite hand shown

NOTES: (1) Mirror of VA-BB-03 (generated). Bend directions unchanged.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-14

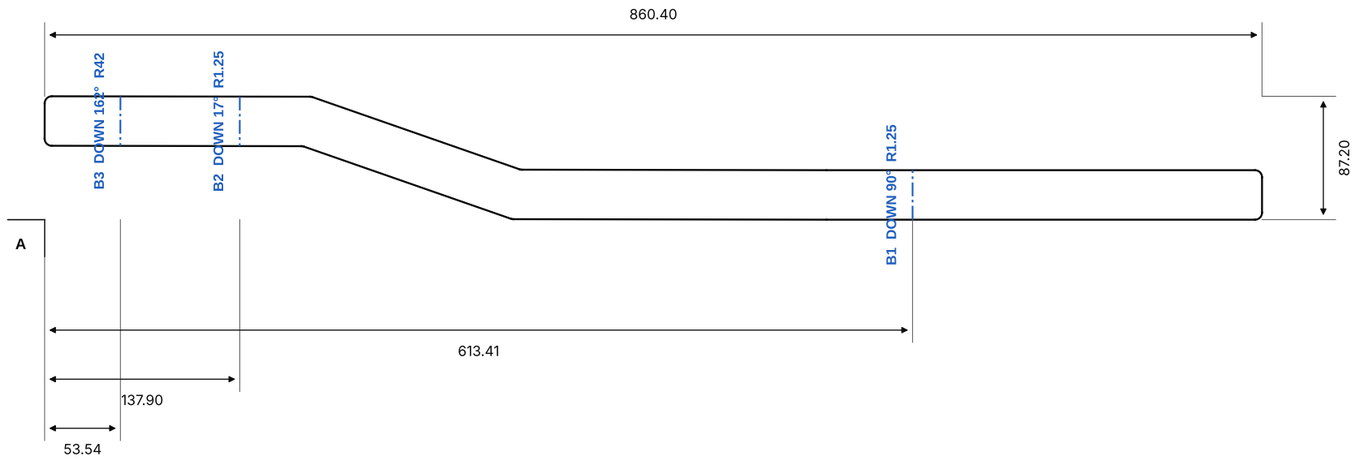
TITLE
Upright LH - FLAT PATTERN

MATERIAL **304 SS 2B**	THICKNESS **5 mm**	QTY / ASSY **1**
FLAT SIZE 306.5 × 768.9	NET MASS 4.66 kg	K-FACTOR 0.46 (air bend, 304)
BENDS 2	SOURCE ITEM #14	DXF VA-BB-14_Upright_LH_5mm.dxf
TOL. LINEAR +/- 0.5 mm	TOL. ANGULAR +/- 1 deg	FINISH Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	1.25	613.41 (X)	3.41	3.09	16
B2	DOWN	17°	1.25	137.90 (X)	0.64	0.33	16
B3	DOWN	162°	42	53.54 (X)	121.35	434.26	roll

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 53.5 / 84.4 / 475.5 / 247.0 mm

FORMED VIEW (reference, not to scale)



NOTES: (1) Bends confirmed by Veld Armour (source sheet 9 note "NO BENDING" is superseded). (2) B3 (162 deg, R42) wraps the R42 top radius of the upright - roll/bump-form it, then fit the strip over the upright edge; B1/B2 on the brake.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-17

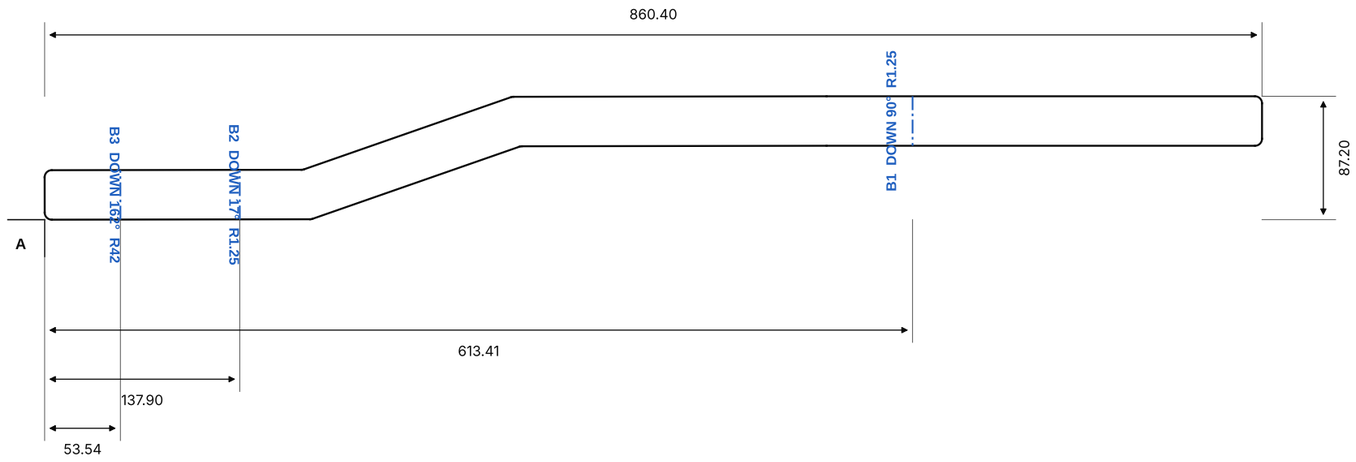
TITLE
Face 1 (RH trim strip) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**2 mm**	**1**
FLAT SIZE	NET MASS	K-FACTOR
860.4 × 87.2	0.48 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
3	#17	VA-BB-17_Face_1_2mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent

FLAT PATTERN - viewed from bend-note face (UP = toward viewer) scale 1:5

red dash-dot = BEND UP blue dash-dot = BEND DOWN green = ETCH/mark datum A = bottom-left corner of flat



BEND TABLE

No	Dir	Angle	R in	From A	BA	BD	V-die
B1	DOWN	90°	1.25	613.41 (X)	3.41	3.09	16
B2	DOWN	17°	1.25	137.90 (X)	0.64	0.33	16
B3	DOWN	162°	42	53.54 (X)	121.35	434.26	roll

BA = bend allowance, BD = bend deduction (mm), K = 0.46, inside radius as listed. V-die = 8T opening (mm).
"From A" = bend-line centre from datum edge. 304: ~1.5x mild-steel tonnage, 2-3 deg extra springback.
WARNING: B1, B2 inside R < 1T - cracking risk in 304; use R >= 2 tooling or prove on first-off.

FORMED SECTION (neutral axis, to scale)



Flats from A: 53.5 / 84.4 / 475.5 / 247.0 mm

FORMED VIEW (reference, not to scale)



opposite hand shown

NOTES: (1) Mirror of VA-BB-17 (generated). Same forming method as VA-BB-17.

VELD ARMOUR

BUILT FOR THE VELD

CNC / PRESS-BRAKE DATA PACK

PART NO.
VA-BB-18

TITLE
Face 1 LH (trim strip) - FLAT PATTERN

MATERIAL	THICKNESS	QTY / ASSY
304 SS 2B	**2 mm**	**1**
FLAT SIZE	NET MASS	K-FACTOR
860.4 × 87.2	0.48 kg	0.46 (air bend, 304)
BENDS	SOURCE ITEM	DXF
3	#18	VA-BB-18_Face_1_LH_2mm.dxf
TOL. LINEAR	TOL. ANGULAR	FINISH
+/- 0.5 mm	+/- 1 deg	Deburr; brush; passivate

Veld Armour (Pty) Ltd - proprietary; not for distribution without written consent